



Faculty of Computer Science

Network Science

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Social Network Analysis of Amherst College Facebook Connections

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Project Overview

Objective:

- Analyze the Amherst College Facebook network (2,235 nodes, 90,954 edges)
- Identify communities and key influencers

Methods:

- Community detection (Louvain, Label Propagation, Girvan-Newman)
- Centrality measures (Degree, Betweenness, Eigenvector)

Tools: Python (NetworkX, Pandas, Matplotlib)

Network Properties

Network Summary:

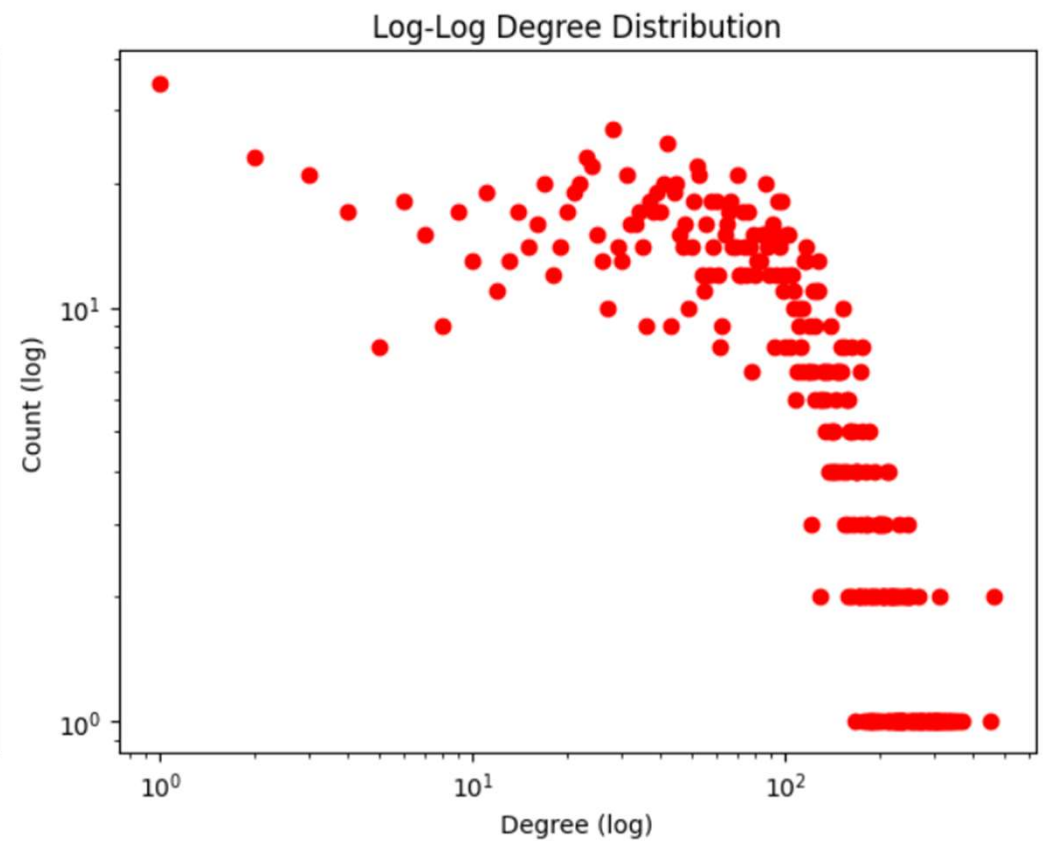
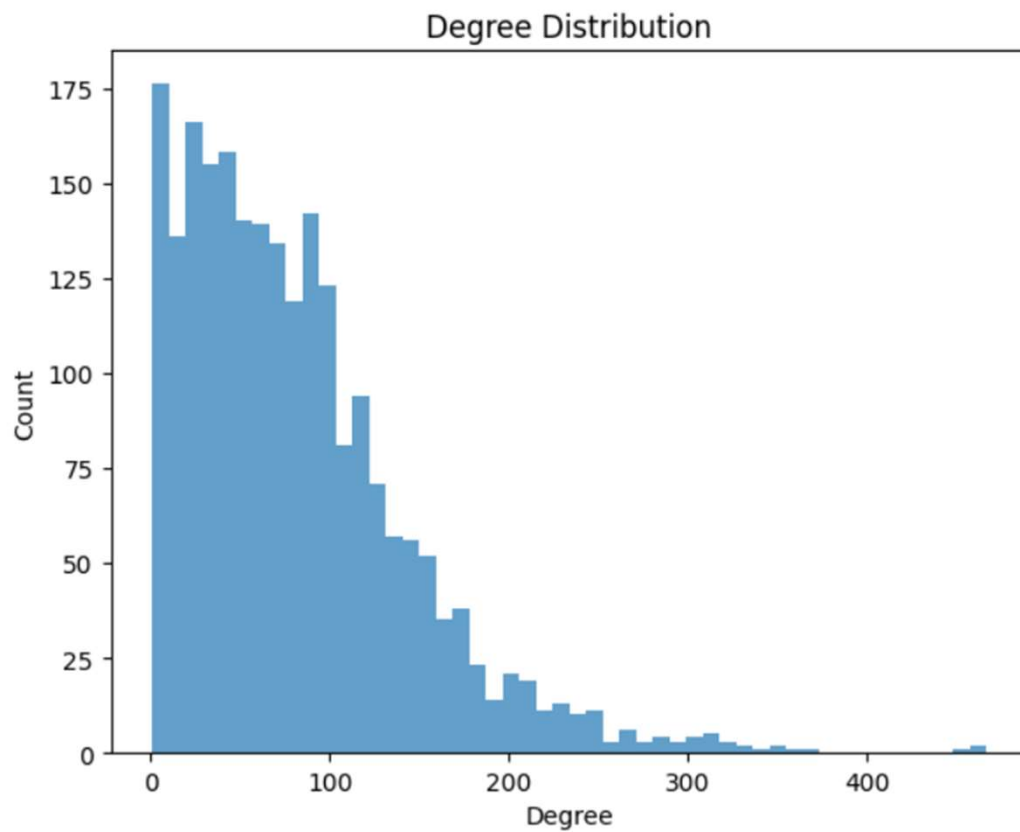
- Edges: 90954 undirected friendships
- Network Type: Undirected, Homogeneous, Unweighted
- Average Degree: 81.39
- Network Density: 0.036433
- Directed/Undirected: Undirected (Facebook friendships are mutual).
- Heterogeneous/Homogeneous: Homogeneous
- Weighted/Unweighted: Unweighted

Degree Distribution

Key Metrics:

Avg. Degree: 81.39 | Clustering: 0.45 (12.5× random)

Avg. Path Length: 2.2 (Small-world effect)



Average Distance & Small-World Property

- Average Path Length (LCC): ~ 2.20 (any two users are ~ 2.2 steps apart on average).
- The network exhibits small-world properties—high clustering (friends of friends tend to know each other) and short path lengths (any two people are connected via a few intermediaries).
- Observed Clustering Coefficient: ~ 0.45 (45% chance that two friends of a user are also friends).
- The actual clustering is $\sim 12.5\times$ higher than in a random network.
- This high clustering is typical of social networks (people form tightly knit friend groups).

Structural Analysis

- The Amherst network is a hybrid of BA and WS:
- BA-like: Scale-free hubs (e.g., popular individuals).
- WS-like: High clustering (friend groups).
- Neither model alone captures both properties perfectly.

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Thank You!